1. ISO 10487 CONNECTOR A (power) + ALWAYS-ON 5V FOR PICO AND GNSS

J1 pad n = ISO A pin n. A1 REVERSE A2 HANDBRAKE A3 AMP REMOTE A4 BATT+ A5 remote (R57) A6 ILLUM A7 ACC A8 GND

Always-on 5.2V: MCU VSYS + GNSS LDO. TPS560430X runs PFM at light load (~55uA Iq).

2. VEHICLE INPUTS (opto-isolated). ACC, ILLUM, REVERSE: +12V = on. PARK: switch to ground = on.

ACC (ISO A7)

ILLUM (ISO A6)

REVERSE lamp (ISO A1)

PARK (ISO A2, handbrake switch; may stay unwired)

LED ~2.7mA at 14.4V; EL817 C-grade CTR >= 200%. 4.7k pull-downs satisfy RP2350-E9.

3. AMPLIFIER REMOTE (+12V from ACC, MCU GP9)

Goes to ISO A3 and C1-6 (A5 via R57). ~1A capable; remote inputs draw <100mA.

Only works while ACC is on, even if the MCU hangs.

4. MCU HELPERS (MCU core is section 13)

GPIO map: see section 13 and DESIGN_rev03.md.

I2S series resistors at the ribbon entry.

Car radio peripheral board rev 0.4 – MG F (1996), VIM3 + RP2350B, 4ch line out on the audio module, NEO-M9N. Generated by gen4.py; see DESIGN_rev04.md.

5. GNSS MODULE: u-blox NEO-M9N, active antenna on U.FL (pigtail to SMA), UART + USB to the main board. Switched 3.3V supply stays on the main board.

6. VIM3 POWER KEY, BACKLIGHT, LIGHT SENSOR (all from the MCU)

Display buttons (to GND): GP16 screen, GP17 vol+, GP22 vol−, GP27 mute. J12.

Backlight and reverse moved off the VIM3 header; the VHAL now sets/reads them through the MCU.

8. VIM3 40-PIN HEADER + ISO 10487 C1 (line out)

J8 pad n = ISO C1 pin n: 1 RL, 2 RR, 3 FL, 4 FR, 5 line GND, 6 remote +12V.

VIM3 pin v -> IDC pad 2v-1 (v<=20) or 2(v-20) (v>20). Pin 1 sits opposite pin 21 (confirmed by Glenn).

Pins 30 (TDM-B MCLK) and 35 (TDM-B lane 3) go straight to the audio module socket J23 as spares.

9. DISPLAY: Waveshare 7in 70H-1024600 adapter built in (HDMI in from VIM3 -> 40-pin FPC to the panel)

Panel 5V (~0.45A) comes from the board's switched 5V rail (+5V_SYS, section 10), not from VIM3 USB. Touch USB goes to the on-board hub (section 11).

Backlight EN/PWM now go straight to the panel: 0V = brightest, 3.3V = darkest (invert PWM in firmware). J6 removed.

FPC orientation as on the Waveshare adapter: seen from the top, ribbon leaving the board edge, pin 1 is on the RIGHT, pin 40 on the left.

10. 5V/5A SYSTEM SUPPLY FOR VIM3 + SCREEN + HUB (TPS55288 buck-boost, rides through cranking and overvoltage)

Input 3V-36V (runs through cranking); output 5.0V up to 5A. MCU GP28 = EN, MCU I2C0 sets OE/voltage/current limit.

VIN is permanent +12V so the VIM3 can finish its Android shutdown after ACC drops.

11. USB HUB: VIM3 header USB (pins 3/4) -> MCU, screen touch, spare port

Hub 1 runs from the always-on +5V_AON so a blank MCU enumerates (first flash over USB). One link to the VIM3: no USB cables. Port 1 MCU, 2 touch, 3 GNSS module USB, 4 second hub (external ports + RTL-SDR). Route D+/D− as 90 ohm pairs.

VIM3 header pins 3/4 are a working USB host port (confirmed by Glenn, 2026-09-29).

13. MCU: RP2350B (QFN-80) + 16MB QSPI flash + 12MHz crystal + 3.3V LDO, per the Raspberry Pi Pico 2 reference design

GPIO0–29 as rev 0.2 except: GPIO40 battery ADC (was GP26); GPIO23/24 USB port faults; GPIO25 LED; GPIO26 USB supply PG.

GPIO29 GNSS safeboot; GPIO30/31 on J10; GPIO32–39 on J21; GPIO41 5V rail ADC; GPIO42 board temperature; GPIO43 hub 1 reset.

RP2350-E9 check (every MCU pin): opto inputs GP2/3/6/7 4.7k down; SBC sense and service jumper 4.7k down; buttons, USB faults/PG, I2C, RUN pulled up;

GNSS PPS/RX 4.7k down (R94/R95); outputs have defined off-state resistors; ADC pins run with the digital input disabled; unused and expansion pins: see DESIGN_rev03.md.

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File: carradio_peripheral_rev04.kicad_sch

Title: Car radio peripheral board (RP2350B + VIM3)

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